



6 – MGF, Discrete Bivariate Dist. and midterms preparations

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Recap: MGF

Let X be a random variable. Its **moment generating function (MGF)** is defined by

$$M_X(t) = \mathbb{E}[e^{tX}], \quad t \in \mathbb{R}.$$

Theorem 3 (Properties of MGF – Part I)

Let X, Y be random variables with $M_X(t) < \infty, M_Y(t) < \infty$ for $-h < t < h$. Then

- (a) Let $M_X^{(n)}(t) = \frac{d^n}{dt^n} M_X(t)$ be the n -th derivative of $M_X(t)$. Then $\mathbb{E}[X^n] = M_X^{(n)}(0)$.
- (b) (Inversion Theorem) If $M_X(t) = M_Y(t)$ for all t , then X and Y have the **same** distribution, i.e. they have the same CDF/PDF.

Recap: MGF

Let X be a random variable. Its **moment generating function (MGF)** is defined by

$$M_X(t) = \mathbb{E}[e^{tX}], \quad t \in \mathbb{R}.$$

Theorem 4 (Properties of MGF – Part II)

(c) If $Y = aX + b$, where $a, b \in \mathbb{R}$, then

$$M_Y(t) = e^{tb} M_X(at)$$

(d) If X and Y are independent, then

$$M_{X+Y}(t) = M_X(t)M_Y(t).$$

Recap: Common Dist.

MGFs of common distributions.

Distribution	MGF
$Bernoulli(p)$	$pe^t + 1 - p$
$Geom(p)$	$\frac{pe^t}{1 - (1 - p)e^t}$ for $t < -\ln(1 - p)$
$Binomial(n, p)$	$(pe^t + 1 - p)^n$
$Poisson(\lambda)$	$e^{\lambda(e^t - 1)}$
$U(a, b)$	$\frac{e^{tb} - e^{ta}}{t(b - a)}$ for $t \neq 0$, 1 for $t = 0$
$N(\mu, \sigma^2)$	$e^{\mu t + \sigma^2 t^2 / 2}$
$Gamma(\alpha, \theta)$	$(1 - \theta t)^{-\alpha}$ for $t < \frac{1}{\theta}$
$Exp(\theta)$	$(1 - \theta t)^{-1}$ for $t < \frac{1}{\theta}$

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Recap: Common Dist.

Discrete Distribution	pmf $\mathbb{P}(X = x)$	$\mathbb{E}[X]$	$\text{Var}[X]$
Bernoulli(p)	$\begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \\ 0 & \text{otherwise.} \end{cases}$	p	$p(1 - p)$
Binomial(n, p)	$\binom{n}{x} p^x (1 - p)^{n-x}, 0 \leq x \leq n$	np	$np(1 - p)$
Geometric(p)	$(1 - p)^{x-1} p, \quad x \geq 1$	$\frac{1}{p}$	$\frac{1 - p}{p^2}$
Poisson(λ)	$e^{-\lambda} \frac{\lambda^x}{x!}, \quad x \geq 0$	λ	λ

Continuous RV	density / pdf $f(x)$	CDF $F(x)$	$\mathbb{E}[X]$	$\text{Var}[X]$
Uniform(a, b)	$\begin{cases} \frac{1}{b-a} & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases}$	$\begin{cases} \frac{x-a}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{if } x < a \end{cases}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Expo(θ), $\theta > 0$	$\frac{1}{\theta} e^{-x/\theta}, \quad 0 \leq x < \infty$	$\begin{cases} 1 - e^{-x/\theta} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$	$\theta = \frac{1}{\lambda}$	$\theta^2 = \frac{1}{\lambda^2}$
Normal(μ, σ^2)	$\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, x \in \mathbb{R}$	Use $Z = \frac{X-\mu}{\sigma}$	μ	σ^2
$Z \sim N(0, 1)$	$\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, x \in \mathbb{R}$	Tables	0	1
Gamma(α, θ) $\Gamma(\alpha) = (\alpha - 1)!$	$\frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-x/\theta}, \quad 0 \leq x < \infty$	N/A	$\alpha\theta = \frac{\alpha}{\lambda}$	$\alpha\theta^2 = \frac{\alpha}{\lambda^2}$
$\chi^2(r)$ or Gamma($\frac{r}{2}, 2$)	$\frac{1}{\Gamma(r/2)2^{r/2}} x^{\frac{r}{2}-1} e^{-x/2}, \quad x \geq 0$	χ^2 Tables	r	$2r$

Refer the graphs of the pdfs in the lecture notes.

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Problem 1 (MGF) (a) Let X be a discrete random variable with PMF p given by

$$\begin{array}{c|c|c|c} x & -1 & 0 & 1 \\ \hline p(x) & \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \end{array} \text{ and } p(x) = 0 \text{ for } x \notin \{-1, 0, 1\}.$$

Compute the MGF of X and use the properties of MGFs from the lecture to compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$.

(b) Let X be a continuous random variable with PDF f given as follows.

$$f(x) = 2x \text{ for } 0 \leq x \leq 1 \text{ and } f(x) = 0 \text{ otherwise.}$$

Compute the MGF $\mathbb{E}[e^{tX}]$ of X for $t \neq 0$.

(c) Let X be a random variable with $M_X(t) = \frac{e^t}{2 - e^t}$ for all $t < \ln 2$. What is the distribution of X ? Hint: Check the table for MGF of common distributions.

Problem 2 (MGF) Let $X \sim \text{Poisson}(\lambda)$. Using the definition of MGF, verify that the MGF of X is given by

$$e^{\lambda(e^t - 1)}.$$

Hence, find the mean and variance of X .

Hint: You may assume that $e^z = \sum_{x=0}^{\infty} \frac{z^x}{x!}$. This is the Maclaurin series for e^z .

Poisson(λ)	$e^{-\lambda} \frac{\lambda^x}{x!}, \quad x \geq 0$	λ	λ
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Problem 3 (MGF) Suppose X and Y are independent Poisson random variables with means λ_1 and λ_2 respectively. What is the distribution of $X + Y$?

Fun fact:

This is called **superposition** in Stochastic Processes.

Recap: Bivariate Dist.

Let X, Y be discrete random variables.

- The **joint probability mass distribution (joint PMF)** of X and Y is defined by

$$p(x, y) = \mathbb{P}(X = x, Y = y).$$

- The **joint cumulative density function (joint CDF)** of X and Y is defined by

$$F(x, y) = \mathbb{P}(X \leq x, Y \leq y) = \sum_{s \leq x} \sum_{t \leq y} p(s, t).$$

Recap: Bivariate Dist.

Let X and Y have the joint probability mass function $f(x, y)$.

- The probability mass function of X alone, which is called the **marginal probability mass function** of X , is defined by

$$p_X(x) = \sum_y p(x, y) = \mathbb{P}(X = x).$$

$p_X(x)$, $p_Y(y)$ are the called the marginal PMF of X and Y .

- If $u(X, Y)$ is a function of X and Y , then

$$\mathbb{E}[u(X, Y)] = \sum_x \sum_y u(x, y) p(x, y)$$

is the **expected value** of $u(X, Y)$.

Recap: Bivariate Dist.

Theorem 9 (Independence via marginals)

*The random variables X and Y are **independent** if and only if*

$$p(x, y) = p_X(x)p_Y(y) \quad \text{for all } x, y.$$

*Otherwise, X and Y are said to be **dependent**.*

Problem 4 (Joint PMF, CDF, Marginal PMF) Let the joint PMF of X and Y be defined by

$$p(x, y) = \frac{x + y}{32},$$

for $x = 1, 2, y = 1, 2, 3, 4$.

- (a) Find $p_X(x)$, the marginal PMF of X .
- (b) Find $p_Y(y)$, the marginal PMF of Y .
- (c) Find $\mathbb{P}(X > Y)$
- (d) Find $\mathbb{P}(Y = 2X)$
- (e) Find $\mathbb{P}(X + Y = 3)$.
- (f) Find $\mathbb{P}(X \leq 3 - Y)$.
- (g) Are X and Y independent or dependent? Why or why not?

Problem 5 (Joint PMF, CDF, Marginal PMF) From a standard poker deck of 52 cards, 3 cards are chosen randomly. Let X be number of clubs among the 3 cards and let Y be the number of hearts among the 3.

- (a) Find the joint PMF of X and Y .
- (b) Find the marginal PMFs of X and Y .
- (c) Are X and Y independent?

$x \setminus y$	0	1	2	3
0	$\frac{200}{1700}$	$\frac{325}{1700}$	$\frac{156}{1700}$	$\frac{22}{1700}$
1	$\frac{325}{1700}$	$\frac{338}{1700}$	$\frac{78}{1700}$	0
2	$\frac{156}{1700}$	$\frac{78}{1700}$	0	0
3	$\frac{22}{1700}$	0	0	0

Problem 6 (Joint PMF, CDF, Marginal PMF) There are eight similar chips in a bowl: three marked $(0, 0)$, two marked $(1, 0)$, two marked $(0, 1)$, and one marked $(1, 1)$. A player selects a chip at random and is given the sum of the two coordinates in dollars. What is the expected payoff?

Bonus Qn: Midterms AY24/25 Qn 6

6. [6 marks] Suppose the stock price today is \$100. Consider the following model for the stock price S tomorrow: A fair coin is tossed 4 times independently. If there are exactly x heads and y tails in the outcome of the 4 tosses, then the stock price tomorrow will be $S = \$100 \times 1.1^x \times 0.8^y$, where $x + y = 4$.

- (a) What is the probability that $S > 90$?
- (b) Compute the variance $\text{Var}[Y]$, where $Y = \ln S$.